Enterprise Lead Management Master Blueprint
Part 8 – DevOps, Infrastructure & Observability

Purpose

Define the infrastructure, deployment, CI/CD, monitoring and operational practices required to run the platform reliably in production.

Deployment Architecture

Deploy React frontend, FastAPI backend, Pipeline Orchestrator, Databricks Lakehouse, WhatsApp integrations and notification services in separate, scalable components behind an API gateway and load balancer.

Environment Strategy

Maintain isolated Development, QA, Staging and Production environments with independent configuration, secrets and deployment pipelines.

Containerization

Package backend services with Docker. Use Kubernetes (or a managed container platform) for scaling, rolling updates and high availability.

CI/CD Pipeline

Implement Git-based workflows with automated linting, unit tests, integration tests, security scans, build, deployment approval, staging validation and production rollout.

Infrastructure as Code

Provision infrastructure using Infrastructure as Code tools. Keep networking, secrets, compute resources and environment configuration under version control.

Monitoring & Observability

Collect application metrics, infrastructure metrics, API latency, Databricks job status, workflow execution, notification delivery and dashboard performance. Create operational dashboards and alerting.

Logging

Centralize structured logs for backend services, pipeline executions, AI decisions, authentication events and integration failures. Support searchable audit logs.

Backup & Disaster Recovery

Define backup schedules, recovery point objectives (RPO), recovery time objectives (RTO), failover procedures, database recovery and periodic restore testing.

Performance & Scalability

Support horizontal scaling, asynchronous processing, queue-based workloads, caching, API rate limiting, autoscaling and performance benchmarking.

Production Readiness Checklist

Validate security, monitoring, backups, alerting, documentation, runbooks, rollback plans, user acceptance testing, load testing and go-live approvals before production deployment.